

Case Study: Call Center Demand Analysis Using Poisson & Monte Carlo Simulation

Problem Statement

Call centers often face fluctuating demand, making it difficult to allocate the right number of agents at the right time. Understaffing leads to long wait times and poor customer experience, while overstaffing increases operational costs.

The goal of this project was to:

- Model hourly call arrivals
 - Understand demand variability
 - Simulate future call volumes
 - Provide a data-driven approach to staffing decisions
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Objectives

- Estimate average call arrival rate (λ)
 - Analyze call distribution patterns
 - Simulate call volumes using Monte Carlo methods
 - Quantify peak demand scenarios
 - Recommend optimal staffing levels
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Tools & Technologies

- R (statistical computing)
 - Poisson distribution modeling
 - Monte Carlo simulation
 - Data visualization (ggplot2)
 - Descriptive statistics
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Approach

1. Data Simulation

Since real call center data was not available, hourly call counts were simulated using a Poisson distribution with an average rate (λ) of 5 calls per hour over a 30-day period (720 observations).

2. Exploratory Data Analysis

- Calculated the mean and variance of call counts
 - Verified that the variance was approximately equal to the mean, supporting the Poisson assumption
 - Visualized the distribution of call volumes to understand frequency patterns
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3. Poisson Modeling

- Estimated the call arrival rate (λ) from the dataset
 - Used the Poisson probability mass function to compute the likelihood of specific call volumes
 - Modeled call arrivals as a random process occurring independently over time
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4. Monte Carlo Simulation

- Generated 10,000 simulated scenarios of future call volumes
 - Used simulation to approximate the distribution of possible outcomes
 - Estimated extreme values (e.g., peak demand conditions) using quantiles
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5. Staffing Optimization

- Determined the 95th percentile of simulated call volumes
 - Estimated the number of agents required based on handling capacity per agent
 - Converted statistical outputs into actionable staffing recommendations
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Key Insights

- Call arrivals closely followed a Poisson distribution, validating the modeling approach
 - The average number of calls per hour remained stable around λ
 - However, variability in call volume highlighted the need for flexible staffing
 - Peak demand (95th percentile) was significantly higher than the average, indicating potential overload periods
 - Monte Carlo simulation provided a practical way to anticipate uncertainty in demand
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Business Impact

- Enabled data-driven workforce planning
 - Helped estimate peak operational load
 - Reduced risk of understaffing during high-demand periods
 - Demonstrated how statistical modeling can improve operational efficiency
 - Provided a scalable framework applicable to real-world call center environments
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Conclusion

This project demonstrates how probability distributions and simulation techniques can be applied to real-world operational problems. By combining Poisson modeling with Monte Carlo simulation, it is possible to better understand uncertainty, forecast demand, and make informed staffing decisions.

Potential Improvements

- Incorporate real call center datasets for validation
 - Extend analysis using time-series forecasting (ARIMA or machine learning models)
 - Build an interactive dashboard for real-time monitoring
 - Integrate with live data pipelines for continuous prediction
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Skills Demonstrated

- Statistical modeling
 - Probability distributions (Poisson, Binomial)
 - Monte Carlo simulation
 - Data analysis in R
 - Data visualization
 - Business problem solving
 - Translating data into actionable insights
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